

Containerization and Orchestration Report Using Docker and Kubernetes

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ayushi2323@LAPTOP-6LVIDNI x + v
ayushi2323@my-nginx:1.0 05b8cb60c354 253MB 69.2MB U
ubuntu:latest 2260313b31c8 160MB 45.3MB U
ayushi2323@LAPTOP-6LVIDNNHT:~/container-k8s-report/container-k8s-report$ doc
ker run -d -p 8080:80 --name k8s-demo-container k8s-demo:1.0
Unable to find image 'k8s-demo:1.0' locally
docker: Error response from daemon: pull access denied for k8s-demo, reposit
ory does not exist or may require 'docker login'

Run 'docker run --help' for more information
ayushi2323@LAPTOP-6LVIDNNHT:~/container-k8s-report/container-k8s-report$ doc
ker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
ayushi2323@LAPTOP-6LVIDNNHT:~/container-k8s-report/container-k8s-report$ doc
ker stop k8s-demo-container

Error response from daemon: No such container: k8s-demo-container
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ayushi2323@LAPTOP-6LVIDNNHT:~/container-k8s-report/container-k8s-report$ dep
loyment.yaml
deployment.yaml: command not found
ayushi2323@LAPTOP-6LVIDNNHT:~/container-k8s-report/container-k8s-report$ kub
ectl get nodes
NAME STATUS ROLES AGE VERSION
desktop-control-plane Ready control-plane 8h v1.36.1
ayushi2323@LAPTOP-6LVIDNNHT:~/container-k8s-report/container-k8s-report$ kub
ectl apply -f deployment.yaml
error: the path "deployment.yaml" does not exist
ayushi2323@LAPTOP-6LVIDNNHT:~/container-k8s-report/container-k8s-report$ kub
ectl get pods
NAME READY STATUS RESTARTS AGE
myapp-9e1d8a1c93-7jq25 1/1 Running 1 (2m37s ago) 7h1m
myapp-9e1d8a1c93-88w7v 1/1 Running 1 (2m37s ago) 7h1m
myapp-9e1d8a1c93-gsg5t 1/1 Running 1 (2m37s ago) 7h1m
myapp-9e1d8a1c93-qdznt 1/1 Running 1 (2m37s ago) 7h1m
myapp-9e1d8a1c93-s9m8g 1/1 Running 1 (2m37s ago) 7h1m
nginx-deployment-7f9e8354c7-5cgx5 1/1 Running 1 (2m37s ago) 7h33m
nginx-deployment-59c4c87bc6-gf4wj 1/1 Running 1 (2m37s ago) 7h33m
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ker ps
CONTAINER ID   IMAGE          COMMAND          CREATED    STATUS    PORTS    NAMES
ayushi2323@LAPTOP-6LVIDNNHT:~/container-k8s k8s-demo-container
ayushi2323@LAPTOP-6LVIDNNHT:~/container-k8s-report/container-k8s-report$ doc
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loyment.yaml
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ayushi2323@LAPTOP-6LVIDNNHT:~/container-k8s-report/container-k8s-report$ kub
ectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
desktop-control-plane              Ready    control-plane   8h    v1.36.1
ayushi2323@LAPTOP-6LVIDNNHT:~/container-k8s-report/container-k8s-report$ kub
ectl apply -f deployment.yaml
error: the path "deployment.yaml" does not exist
ayushi2323@LAPTOP-6LVIDNNHT:~/container-k8s-report/container-k8s-report$ kub
ectl get pods
NAME                                READY    STATUS    RESTARTS    AGE
myapp-9e1d8a1c93-7jq25              1/1      Running    1 (2m37s ago)  7h1m
myapp-9e1d8a1c93-88w7v              1/1      Running    1 (2m37s ago)  7h1m
myapp-9e1d8a1c93-gsg5t              1/1      Running    1 (2m37s ago)  7h1m
myapp-9e1d8a1c93-qdznt              1/1      Running    1 (2m37s ago)  7h1m
myapp-9e1d8a1c93-s9m8g              1/1      Running    1 (2m37s ago)  7h1m
nginx-deployment-7f9e8354c7-5cgx5   1/1      Running    1 (2m37s ago)  7h33m
nginx-deployment-59c4c87bc6-gf4wj   1/1      Running    1 (2m37s ago)  7h33m
```

1. Objective

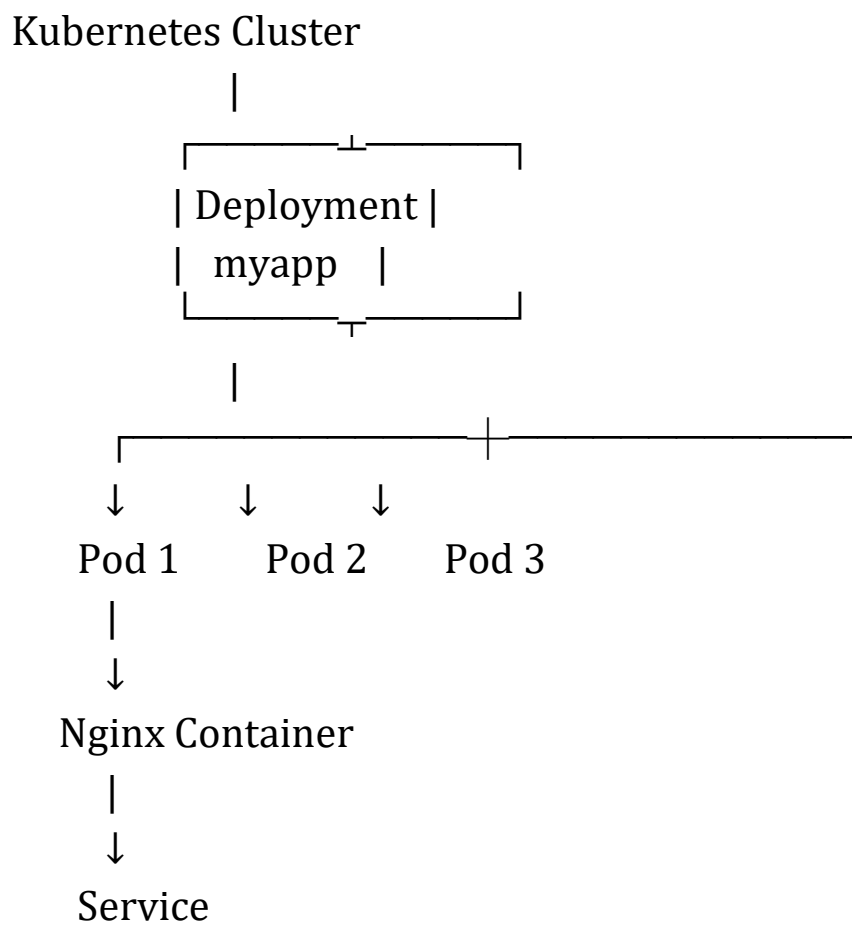
The objective of this practical is to containerize a web application using Docker and deploy it on a Kubernetes cluster. The practical demonstrates container creation, Kubernetes deployment, service exposure, scaling, and verification of the application.

2. Tools and Technologies

Tool	Purpose
Docker	Containerization
Kubernetes	Container orchestration

Tool	Purpose
kubectl	Kubernetes cluster management
Nginx	Web server
HTML	Sample application

3. Architecture



4. Containerization

Explain:

1. Created index.html.
2. Created Dockerfile.
3. Built Docker image.
4. Started a Docker container.
5. Tested the application through the browser.

Insert **Screenshots 1–5**.

5. Kubernetes Deployment

Explain:

1. Created deployment.yaml.
2. Created service.yaml.
3. Applied both manifests using kubectl.
4. Verified Kubernetes nodes and pods.
5. Accessed the application through the Kubernetes Service.

Insert **Screenshots 6–11**.

6. Scaling Strategy

Write:

The Kubernetes Deployment was initially configured with two replicas. The application was then scaled to four replicas using the kubectl scale command. Kubernetes automatically created additional pods to achieve the desired replica count. The application was subsequently scaled down to two replicas.

Add a small table:

Stage	Replicas Result	
Initial deployment	2	2 pods running
Scale-up	4	4 pods running
Scale-down	2	2 pods running

Insert **Screenshots 12–14**.

7. Observations

You can write:

The Docker image successfully packaged the web application and its required web server environment. Kubernetes successfully deployed the application using a Deployment and exposed it through a Service. Multiple replicas of the application were maintained by Kubernetes. Scaling the Deployment increased or decreased the number of running pods according to the desired replica count. The application remained accessible through the Kubernetes Service during the deployment and scaling operations.

8. Result

The web application was successfully containerized using Docker and deployed on a Kubernetes cluster. The application was verified through running pods and a Kubernetes Service. Scaling operations were successfully performed from two replicas to four replicas and back to two replicas.

9. Conclusion

This practical demonstrated the complete containerization and orchestration workflow using Docker and Kubernetes. Docker was used to package the application into a container image, while Kubernetes was used to deploy, manage, expose, and scale the application. The experiment demonstrated the benefits of orchestration, including replica management, service exposure, and scalable application deployment.